IVAD: Evidence-Constrained and Risk-Controlled Failure Diagnosis for AI Agents

Hanzhe Liu
School of Computer Science and Engineering, Sun Yat-sen University
Guangzhou, China

Abstract

Diagnosing failures in tool-using agents requires more than locating a suspicious step: a trustworthy report must identify a causal failure, bind each claim to re-addressable trajectory evidence, and expose when that evidence is insufficient. Existing attribution and critic pipelines can produce plausible reports, yet an authentic citation may be irrelevant and a second model may repeat the diagnoser's error. We introduce IVAD, Independently Verified Agent Diagnosis, which treats diagnosis as an evidence-constrained selective decision. IVAD combines a versioned claim–evidence contract, deterministic integrity and scope checks, a semantically separated verifier, one bounded revision, and explicit abstention or human review. It further specifies a group-selective risk protocol that applies simultaneous exact-binomial bounds over a frozen finite candidate family and evaluates the selected operating point once on untouched test data. We realize IVAD in SpanVouch, an open-source system with six public contract roots, recoverable review state, framework and provider adapters, and manifest-bound evaluation artifacts. System validation spans SupportLab and OpsLab through LangGraph and AutoGen, exercises 24 domain–framework–condition cells offline, and verifies the complete artifact pipeline without provider or GPU calls. These results establish protocol executability and reproducibility; empirical semantic-verification gains and target-risk attainment remain separate, artifact-gated claims.

Keywords

AI agents, failure diagnosis, independent verification, evidence grounding, selective risk control

1 Introduction

Tool-using agents interleave model decisions with retrieval, API calls, code execution, and interactions among agents. When an execution fails, the final error rarely identifies the decisive cause. A malformed argument may be repaired downstream, an early context error may surface only after several valid actions, and a failed tool call may be a consequence rather than the cause. Agent failure diagnosis must therefore recover a causal report from the trajectory and ground that report in evidence that another process can locate and check.

Recent systems make trajectory-based attribution increasingly concrete. AgentRx derives and checks execution constraints before localizing a critical step [3]; AgenTracer combines controlled injection, counterfactual replay, and learned attribution [23]; and VerifyMAS verifies failure hypotheses over complete multi-agent trajectories [17]. Who&When Pro further scales controlled failure injection and golden attribution labels [13]. These methods improve localization, supervision, and hypothesis testing, but they leave a downstream decision unresolved: when should a system trust a structured diagnosis rather than treat it as another plausible model output?

Three failure modes make this trust decision distinct from attribution alone. First, citation validity is weaker than evidential support: a model can cite a real field that is irrelevant, insufficient, or contradicted elsewhere in the trajectory. Second, a critic does not create independent evidence. A diagnoser and verifier can share a model family, provider, prompt assumptions, retrieved summaries, or the same misleading trajectory abstraction. Third, deployment requires selective acceptance. Rejecting uncertain reports can lower observed error only by sacrificing acceptance, while a threshold chosen after inspecting calibration outcomes requires selection-aware uncertainty accounting.

We introduce *Independently Verified Agent Diagnosis* (IVAD), a protocol that makes these trust boundaries explicit. Each diagnosis contains a bounded causal chain whose claims resolve to immutable trajectory fields and canonical hashes. A deterministic channel enforces identity, integrity, temporal, structural, scope, and evidence-coverage constraints; a separately controlled semantic channel re-evaluates relevance, sufficiency, counter-evidence, and alternative causal accounts without access to evaluator labels or the diagnoser's hidden reasoning. Verification can accept, request at most one auditable revision, abstain, or defer to a human decision.

IVAD also formalizes acceptance at the level where its risk statement is defined. Reports that share a preregistered task-template, repository, and environment tuple form one group. The protocol freezes the loss, scoring rule, finite candidate family, minimum accepted-group count, and tie-breaker before calibration. Simultaneous one-sided exact-binomial bounds identify candidates whose group-selective risk is at most a target α under independent-group and frozen-pipeline assumptions. The selected candidate is evaluated once on untouched test data; if no candidate is feasible, the protocol returns no operating point rather than relaxing the target.

SpanVouch is the open-source reference implementation of these ideas. It provides six versioned public contracts, strict trace projection, deterministic and optional semantic verification, one bounded revision, human authority, recoverable SQLite state, API and command-line delivery, LangGraph and AutoGen adapters, two controlled agent labs, and content-addressed evaluation artifacts. The checked-in validation traverses the complete 24-cell offline domain–framework–condition matrix with no provider or GPU calls. This evidence tests executability, parity, provenance, and recovery; it does not substitute fake-provider behavior for a semantic-effectiveness result.

Source code: github.com/naturaljam/SpanVouch.

This paper makes four contributions:

- We define IVAD, a diagnosis protocol whose causal claims bind to immutable trajectory fields through a machine-checkable claim–evidence contract.
- We separate hard factual eligibility from controlled semantic verification and make diagnoser–verifier information overlap an explicit experimental variable.
- We formulate accepted-diagnosis correctness as group-selective risk and give a finite-candidate, simultaneous calibration protocol with an explicit no-feasible-point outcome.
- We develop SpanVouch, a reproducible implementation that connects contracts, diagnosis, verification, bounded review, durable recovery, framework adapters, and manifest-bound evaluation in one system.

2 Background and Related Work

2.1 Failure Attribution for LLM Agents

Agent failure attribution has progressed from direct trajectory classification to more structured forms of localization and verification. Who&When formalizes responsible-agent and decisive-step attribution, while TRAIL and MAST provide trajectory corpora and failure taxonomies [4, 7, 24]. AgentRx synthesizes constraints and validates them step by step before an LLM judge identifies a critical step and failure category [3]. AgenTracer uses counterfactual replay and programmed fault injection to train a specialized failure tracer [23]. VerifyMAS changes the inference problem from direct agent–error prediction to full-trajectory hypothesis verification [17]; DoVer uses intervention and checkpoint replay to test debugging hypotheses [15]. HarnessFix further joins trace representation, diagnosis, scoped repair, and regression validation [5]. Who&When Pro contributes a much larger controlled benchmark constructed by replaying a successful prefix before injecting a failure [13]. IVAD is complementary to these localization methods: it studies whether a produced diagnosis should be accepted, acquired upon, or rejected after its evidence and verification failure modes are explicitly represented.

2.2 Uncertainty and Risk Control for Agent Attribution

Conformal Agent Error Attribution constructs contiguous prediction sets over trajectory locations and provides finite-sample coverage guarantees for sequential error localization [8]. Its prediction object is a set of possible error locations used for rollback. Our primary risk object is different: a bounded loss over the correctness and evidential sufficiency of an accepted diagnosis report. Conformal Risk Control provides a general finite-sample framework for bounded monotone losses [1]. Selective Conformal Risk Control combines selective prediction with conformal risk control and includes a calibration-only PAC-style variant [21]. We use this line as the statistical basis for testing risk–coverage control and compare against more general risk-controlled selective trust decisions such as SCoRE [2], while treating exchangeability, simultaneous threshold selection, and acquisition-induced distribution changes as explicit assumptions rather than automatic properties of an agent system.

2.3 Verification and Critic-Based Agent Workflows

Generator–critic and self-refinement workflows commonly use another model call to review or revise an output. For failure diagnosis, however, the number of reviewers is less important than their information and failure overlap. AgentRewardBench shows that trajectory judges have systematic and benchmark-dependent weaknesses [14]; LLM critics can surface bugs but can also introduce false findings [16]; and distinct models may retain strongly correlated errors [11]. GroundEval shows the complementary value of checking stateful evidence paths with deterministic constraints rather than final-answer plausibility alone [9]. IVAD therefore separates deterministic facts from semantic judgments, restricts the semantic verifier to a structured diagnosis and sanitized source trajectory, and measures the effect of sharing models, providers, prompts, and retrieved summaries. This turns "independent verification" from a role name into an experimental variable.

2.4 Adaptive Evidence Acquisition

Adaptive retrieval methods allocate information-gathering effort according to estimated task difficulty or model uncertainty. Adaptive-RAG routes questions among no-retrieval, single-step, and iterative retrieval strategies [10]; DRAGIN decides when and what to retrieve during generation [18]; and SeaKR uses self-aware uncertainty to select knowledge expected to reduce uncertainty [22]. Related agentic search and adaptive elicitation methods collect information sequentially during reasoning [12, 19]. These systems primarily acquire information to improve task answers. IVAD instead acquires provenance-preserving evidence to resolve a verifier-identified diagnostic gap under an explicit action budget, then recalibrates the complete post-acquisition decision policy. BCEA is especially close at the decision level: it replaces a binary answer–abstain choice with answering, abstention, or bounded evidence acquisition for vision–language claims [20]. IVAD instead acts over typed diagnostic evidence gaps, preserves immutable trajectory provenance, and defines risk over accepted diagnosis reports rather than task answers.

Taken together, these lines provide the individual mechanisms on which IVAD builds: trajectory localization, evidence-path checking, critic judgments, selective risk control, and budgeted acquisition. IVAD does not claim any one of these components in isolation. Its distinction is their composition around an evidence-bearing diagnosis contract, controlled verifier contexts, preregistered group-level acceptance, and recalibration of the complete policy after acquisition; SpanVouch makes that composition executable.

3 Problem Formulation

3.1 Agent Trajectories and Addressable Evidence

Let an agent execution be represented by an immutable trajectory $\tau = (s_1, \ldots, s_n)$, where each span s_i records an entity, operation, parent relation, time interval, status, and a finite map of attributes. A stable selector $p = (i, f)$ addresses field f of span s_i. Resolving p against τ yields an evidence item

$$e = (i, f, v, H(v)), \tag{1}$$

where v is the canonical field value and H is SHA-256 over its canonical serialization. An evidence reference is valid only if the span and field exist, the observed value equals v, and the recorded digest equals $H(v)$. Validity establishes that a citation is authentic; it does not establish that the evidence is relevant or sufficient for a causal claim.

3.2 Structured Diagnosis

A diagnosis report is

$$d = (r, y, S, C, E, q, m), \tag{2}$$

where $r \in \{\text{DIAGNOSED}, \text{NO-FAILURE}, \text{ABSTAINED}\}$ is the report status, y is a failure type when $r = \text{DIAGNOSED}$, S is a set of decisive spans, $C = (c_1, \dots, c_k)$ is a bounded causal chain, E is a set of evidence references, $q \in [0, 1]$ is a model-reported confidence, and m is provenance. Each claim c_j names its causal stage and the subset of E used to support it. A report may instead return NO-FAILURE or abstain with a structured reason. Contract v1 does not encode a separate responsible-entity field; attribution is represented by decisive spans and causal claims. Entity-level evaluation is therefore defined only when the reference label is derivable from span metadata or supplied through a separately versioned contract. The contract constrains syntax, identity, integrity, and state consistency; semantic support is evaluated separately.

3.3 Independent Verification Protocol

Given (τ, d), verifier channel k produces

$$z_k = (u_k, F_k, G_k, m_k), \tag{3}$$

where $u_k \in \{\text{VERIFIED}, \text{NEEDS-EVIDENCE}, \text{REVIEW-REQUIRED}\}$, F_k is a set of findings, G_k is a set of typed evidence gaps, and m_k records verifier provenance. Hard deterministic findings—for example, an invalid selector, hash mismatch, impossible time order, or scope violation—make a report ineligible for automatic acceptance. The semantic channel evaluates relevance, sufficiency, counter-evidence, and alternative causal explanations. We call the protocol independently verified when channel inputs and failure sources are deliberately separated and their residual error correlation is measured; we do not assume statistical independence a priori.

3.4 Selective Diagnosis Risk

Let $L^{\text{rep}}(d, y^*) \in \{0, 1\}$ be a report-level loss against an execution-derived reference y^*. The loss is one when any prespecified critical component is wrong: report status, failure type, decisive-span set, causal relation, or evidence sufficiency. Entity attribution may be added only for datasets whose gold responsibility label is derivable from the frozen span representation. Equivalent sufficient-evidence sets and the adjudication procedure are frozen before calibration. Adjudication is completed without access to calibration scores or acceptance decisions, and every sampled report receives a certification loss $\widetilde{L}^{\text{rep}}$: it equals the adjudicated loss when all critical labels resolve and is set conservatively to one if any critical label remains unresolved. No report or group is excluded because a critical label is unresolved.

The independent Bernoulli unit is a preregistered group G_i keyed by the tuple (task template, repository, environment); all report opportunities that share the tuple belong to the same group. These groups are sampled independently from the in-distribution group population. Their report opportunities $j = 1, \dots, N_i$ and grouping keys are fixed without using calibration outcomes. A frozen scalar error score $s(x)$ assigns larger values to riskier reports. At candidate λ, report-level acceptance is

$$B_{ij}(\lambda) = \mathbf{1}\{s(X_{ij}) \le \lambda\}\mathbf{1}\{H_{ij} = 1\}, \tag{4}$$

where H_{ij} is deterministic eligibility. The prespecified group policy accepts a group iff it accepts at least one report and assigns a binary group loss iff any accepted report has a frozen critical defect:

$$A_i^G(\lambda) = \mathbf{1}\left\{\sum_{j=1}^{N_i} B_{ij}(\lambda) > 0\right\}, \tag{5}$$

$$L_i^G(\lambda) = \mathbf{1}\left\{\sum_{j=1}^{N_i} B_{ij}(\lambda)\widetilde{L}_{ij}^{\text{rep}} > 0\right\}. \tag{6}$$

The population group acceptance rate and primary selective risk are

$$a_G(\lambda) = \Pr(A_i^G(\lambda) = 1), \tag{7}$$

$$p_\lambda^G = \Pr(L_i^G(\lambda) = 1 \mid A_i^G(\lambda) = 1). \tag{8}$$

Hard deterministic violations are never eligible, regardless of their learned risk score.

For n independently sampled calibration groups, define

$$M_\lambda^G = \sum_{i=1}^{n} A_i^G(\lambda), \quad K_\lambda^G = \sum_{i=1}^{n} A_i^G(\lambda)L_i^G(\lambda). \tag{9}$$

Empty group acceptance is infeasible, not zero risk, and $K_\lambda^G / M_\lambda^G$ is defined only when $M_\lambda^G > 0$. The analogous individual-report ratio and its report counts are descriptive because reports within a group need not be independent.

Before calibration, we freeze the score, deterministic tie-breaking, minimum accepted-group count $m_{\min}$, and a finite candidate family Λ. For overall failure probability δ, let U_λ be the one-sided Clopper–Pearson upper bound for p_λ^G, using per-candidate failure probability $\delta/|\Lambda|$ so that the bounds hold simultaneously over the family [6]. Explicitly,

$$U_\lambda = \begin{cases} 1, & M_\lambda^G = 0 \text{ or } K_\lambda^G = M_\lambda^G, \\ \mathrm{Beta}^{-1}\left(1 - \dfrac{\delta}{|\Lambda|}; K_\lambda^G + 1, \; M_\lambda^G - K_\lambda^G\right), & \text{otherwise.} \end{cases} \tag{10}$$

The feasible set is

$$\mathcal{F} = \{\lambda \in \Lambda : U_\lambda \le \alpha, \; M_\lambda^G \ge m_{\min}\}. \tag{11}$$

Conditional finite-sample guarantee. For every fixed $\lambda \in \Lambda$, conditional on the number of accepted groups, the erroneous accepted-group count is binomial with parameter p_λ^G under independent group sampling and a frozen pipeline. The one-sided exact bounds therefore satisfy

$$\Pr\Big(\forall \lambda \in \Lambda : p_\lambda^G \le U_\lambda\Big) \ge 1 - \delta \tag{12}$$

by the union bound. Consequently, whenever $\mathcal{F}$ is nonempty, any candidate selected from $\mathcal{F}$ by the frozen rule satisfies $p_\lambda^G \le \alpha$ with probability at least $1-\delta$. This statement is conditional on the defined sampling population, frozen loss, complete candidate family, and absence of calibration-dependent pipeline changes.

The chosen operating point maximizes calibration group acceptance over $\mathcal{F}$, with the frozen tie-breaker, and is evaluated once on untouched test data. If $\mathcal{F}$ is empty, the outcome is "no feasible operating point." This yields a calibration-only PAC certificate under frozen-pipeline and i.i.d.-group assumptions [21]; it is not ordinary conformal risk control applied directly to the random selective-risk ratio [1]. With zero erroneous accepted groups, one fixed candidate, and one-sided 95% confidence, the exact upper bound requires 59 accepted independent calibration groups for $\alpha = 0.05$ and 29 for $\alpha = 0.10$. Alternative treatments of unresolved labels are reported separately as sensitivity analyses and do not inherit this certificate.

Untouched-test evaluation. The selected candidate is fixed before test access and evaluated exactly once. Test counts and uncertainty estimate its realized performance; they neither repeat candidate selection nor retroactively alter the calibration certificate.

3.5 Bounded Evidence Acquisition

When verification yields an actionable gap G, an acquisition policy $\pi(G)$ may choose one action from a finite allowlist. Level 0 uses existing evidence, Level 1 reveals an already recorded but omitted field or span, and Level 2 replays at most one allowlisted read-only and idempotent operation. The action, returned evidence, cost, and resulting diagnosis revision are logged. Because acquisition changes the observed feature distribution, the decision rule after acquisition is calibrated for the complete policy over routing, action, diagnosis revision, re-verification, and final acceptance [20]. If the acquisition policy is selected on calibration data, the simultaneously corrected family is the frozen policy-by-threshold family; otherwise the policy is frozen on development data before calibration. No pre-acquisition threshold is reused after acquisition. If no action is permitted or the post-acquisition report remains ineligible, the system abstains.

4 The IVAD Method

4.1 Overview

IVAD is designed to convert an agent trajectory into a selective, evidence-constrained diagnostic decision. Given a failed execution, the method (1) projects the raw trace into a sanitized diagnostic context and addressable evidence catalog, (2) generates a structured diagnosis, (3) independently re-parses its evidence through deterministic and semantic verification channels, and (4) chooses among acceptance, one bounded acquisition action, and abstention. Figure 1 shows how SpanVouch realizes projection, structured diagnosis, dual-channel verification, bounded revision, human decision, and manifest-bound persistence. The group-selective controller and Level 1/2 acquisition actions are protocol components whose empirical claims require qualifying calibration and provider artifacts; the engineering results reported here do not substitute for those outcomes.

4.2 Claim–Evidence Contract

The claim–evidence contract prevents natural-language explanations from being treated as self-authenticating evidence. Each causal claim names one or more evidence identifiers, and every identifier resolves through a stable selector to an immutable trajectory value and digest. Deterministic validation checks the span tree, selector existence, canonical value, hash, duplicate references, critical-span grounding, scope, and domain invariants. These checks are hard constraints: a semantic score cannot override them.

Contract v1 has six roots: `spanvouch.trace/1.0`; `spanvouch.diagnosis/1.0`; `spanvouch.diagnostic-context/1.0`; `spanvouch.verification/1.0`; `spanvouch.review/1.0`; `spanvouch.artifact-manifest/1.0`. Package versions and contract versions are independent: a package release need not change a contract, and a contract revision is not implied by a package release. Selectors, canonical values, and hashes are therefore machine-checkable bindings. They do not establish that a cited item is semantically relevant or that the cited items are causally sufficient for a claim.

4.3 Controlled Failure Separation

The semantic verifier receives the structured claim, sanitized trajectory, and allowed evidence catalog, but not the diagnoser's hidden reasoning or evaluator labels. It judges whether the cited facts support the claimed failure and responsibility, whether decisive counter-evidence is unaddressed, and whether an alternative causal account remains viable. The experimental design varies model identity, provider, prompt, retrieved summaries, and visible rationale to measure which forms of separation reduce correlated false acceptance. The B0–B5 design compares these conditions across paired LangGraph and AutoGen scenario-template clusters. The offline matrix reported in this paper validates those interfaces with declared fake providers; estimating incremental defect detection or correlated false acceptance requires real-provider outcomes.

4.4 Group-Selective Risk Protocol

Verification produces hard eligibility constraints and soft features such as channel disagreement, evidence coverage, semantic support, and distributional indicators. IVAD freezes a scalar error score, its direction, deterministic tie-breaking, a finite candidate family Λ, target α, confidence level $1 - \delta$, and minimum accepted-group count $m_{\min}$ before calibration. The independently sampled unit is a preregistered task group, not an individual report. A group incurs loss when any accepted report within it has a frozen critical defect.

Calibration records $(K_\lambda^G, M_\lambda^G)$ for every candidate and applies simultaneous one-sided exact-binomial bounds. The protocol selects the feasible candidate with greatest calibration group acceptance, applies a frozen tie-breaker, and evaluates the result once on untouched test data. If no candidate meets both the risk bound and accepted-group minimum, IVAD reports no feasible operating point. Section 3 states the full conditional finite-sample guarantee and its assumptions. SpanVouch supplies the contracts, artifact identities, and evaluation pipeline needed to execute this protocol; the engineering results in this paper do not claim empirical target-risk attainment.

4.5 One-Shot Layered Acquisition

An acquisition action is triggered only by a typed evidence gap. The policy selects the least costly permitted level, records the action and budget, allows at most one diagnosis revision, and then repeats verification. The no- acquisition and post-acquisition paths use a jointly frozen calibration protocol over routing, action, revision, re-verification, and final acceptance. If calibration labels select an acquisition policy, Λ is the complete frozen policy-by-threshold family covered by the simultaneous correction; otherwise the policy is frozen on development data before calibration. No pre-acquisition threshold is reused after acquisition. This design makes the acceptance recovered by acquisition measurable against tokens, tool calls, latency, and monetary cost.

5 SpanVouch Reference Implementation

5.1 Architecture and Contracts

SpanVouch implements IVAD as a Python system with dependency direction `contracts` ← `trace` ← `diagnosis` ← `verification` ← `review`. Delivery, storage, framework, and model integrations remain outside these core layers. Architecture tests enforce the direction so that a provider or orchestration framework cannot redefine the evidence and decision semantics.

The public Contract v1 surface has six roots: TraceIR, diagnostic context, diagnosis, verification, review, and artifact manifest. Models reject unknown fields and use canonical JSON for stable SHA-256 identities. A package release does not implicitly change a contract version, which allows stored traces, reports, and decisions to retain their original interpretation.

5.2 Trace Projection and Evidence Binding

`TraceProjector` converts a raw execution into a sanitized diagnostic context while preserving stable span identity and causal structure. `EvidenceCatalog` resolves selectors of the form `span-id::field-path` to a canonical value and digest. Provider-visible views exclude credentials, authorization headers, raw provider responses, hidden reasoning, evaluator labels, mutation metadata, and split identity.

`DiagnosisEngine` produces a bounded report containing status, failure type, decisive spans, at most three staged causal claims, evidence references, confidence, and provenance. The deterministic rules diagnoser supplies the offline path; a provider-backed diagnoser uses the same output contract and must pass strict parsing before its report enters verification.

5.3 Separated Verification

`DeterministicVerifier` re-resolves every cited selector from the stored context. It checks report binding, canonical values and hashes, duplicate references, causal and critical-span grounding, evidence budgets, temporal and scope constraints, clean-trace conflicts, and domain invariants. These checks form hard eligibility: a semantic score cannot override a failed integrity or scope rule.

`SemanticVerifier` is an optional provider path that receives only the sanitized context and structured diagnosis. It evaluates relevance, sufficiency, counter-evidence, and alternative causal accounts, then returns a strict verification report. Invalid provider output becomes a typed human-review requirement rather than triggering an untracked repair call. The experiment adapters can vary model, provider, prompt, and visible context while holding the diagnosis bytes fixed.

5.4 Bounded Review and Recovery

Verifier reports use three workflow verdicts: VERIFIED, NEEDS-EVIDENCE, and REVIEW-REQUIRED. The review state machine permits at most one evidence-guided diagnosis revision and repeats verification after that revision. Every path terminates in a human CONFIRM, CORRECT, or REJECT decision; automation does not manufacture an authenticated reviewer identity.

SQLite persistence stores sanitized snapshots, revisions, verifier reports, events, and terminal decisions. Idempotency keys, compare-and-swap transitions, owner-fenced leases, and immutable event order make interrupted cases recoverable across processes and service restarts. FastAPI and the HTTP-only command-line client expose the same application workflow. The container image runs as an unprivileged user and stores review state in a persistent volume.

5.5 Framework and Evaluation Adapters

SupportLab and OpsLab create controlled tool-using failures with execution- derived labels. LangGraph and AutoGen adapters implement a shared runtime port; parity checks compare terminal state, tool-call order, failure injection, and TraceIR projection before a trace becomes eligible for paired evaluation. The replay repository then freezes content-addressed traces so that verifier conditions consume identical diagnosis and trajectory bytes.

`ArtifactManifest` binds code, contracts, datasets, configuration, prompts, model and provider identities, randomness, runtime, usage, cost, and payload provenance. The experiment runner fails closed before live execution unless an exact approved manifest, allowlisted plan, budget ledger, and explicit live flag agree. Offline rules, deterministic verification, artifact analysis, and the complete delivery path require no provider credential.

The dashed path in Figure 1 marks IVAD's formal group-selective controller and post-acquisition policy. SpanVouch supplies the contract and artifact interfaces required by that protocol; the results in this paper validate the implemented engineering path rather than empirical target- risk attainment.

6 Evaluation

The evaluation separates properties that can be established without a model provider from claims that require real semantic-verifier observations. We ask three questions. **RQ1** tests whether the implemented evidence contract accepts contract-valid reports and intercepts controlled integrity, grounding, scope, and conflict defects. **RQ2** tests whether SpanVouch preserves determinism, recovery, dependency boundaries, and delivery behavior across its full release gate. **RQ3** tests whether the multi-domain, multi-framework B0–B5 artifact pipeline executes end to end without leaking labels or requiring a live provider. The formal semantic-effectiveness and group-selective-risk study remains governed by the protocol below, but its unexecuted outcomes are not mixed with RQ1–RQ3.

6.1 Frozen Deterministic Benchmark

The primary controlled benchmark contains 36 SupportLab diagnosis candidates derived from 20 frozen traces. Twenty candidates are contract-valid, unmodified reports. The other 16 contain one programmed defect from six families: unsupported scope (6), invalid selector (2), evidence-value hash mismatch (2), ungrounded claim (2),

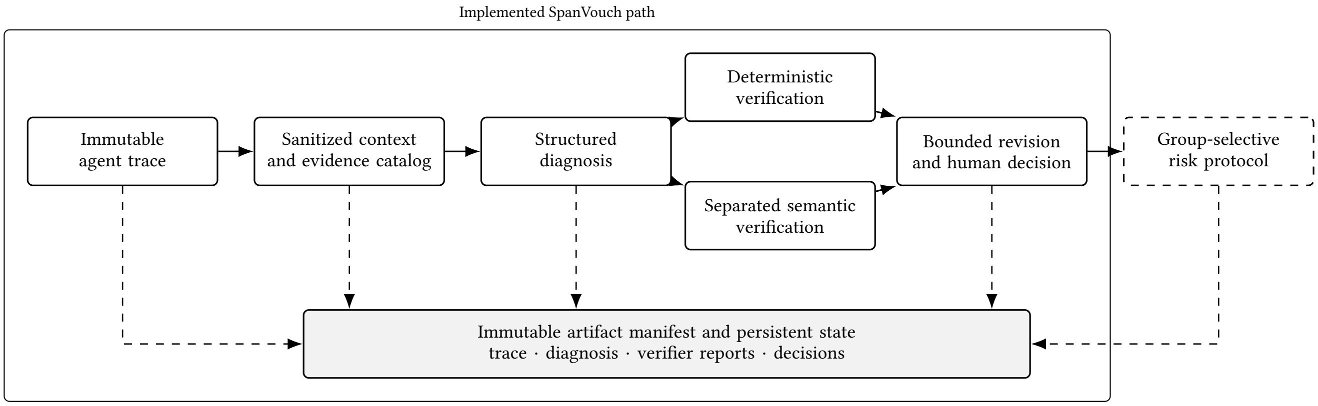

Figure 1: IVAD information and decision flow in SpanVouch. Solid boxes are implemented components exercised by the engineering validation. The dashed box is the formal group-selective protocol; the paper specifies its conditional guarantee but does not report empirical target-risk attainment.

ungrounded critical span (2), and diagnosis conflict (2). The deterministic verifier processes every candidate without a provider call. A valid candidate succeeds only when it receives VERIFIED; a mutation is intercepted when it receives NEEDS-EVIDENCE or REVIEW-REQUIRED with the corresponding typed finding.

The checked-in reference bundle binds the candidate and label manifests, configuration, code revision, runtime, and output hashes. Its artifact-manifest SHA-256 is f8f33da259ff055edf626ac50f351 02e1763e7ee00e8d9a046003e945f1ee3a5; the metrics SHA-256 is c3fc1f4fc2015cbc0a3d6691bf01da98e1a77f7ac139d37f7e75cd2038 742f9b.

6.2 Release and Recovery Validation

The release gate combines unit, contract, architecture, integration, process, and end-to-end tests. Contract tests cover the six public schema roots and valid/invalid fixtures. Architecture tests enforce dependency direction and provider-label isolation. Recovery tests exercise idempotency, compare-and-swap updates, owner-fenced leases, multi-process SQLite access, and state persistence across an API-container restart. Static and delivery gates run Ruff, strict mypy, locked wheel construction, Compose validation, and an unprivileged container smoke test.

6.3 Offline Multi-Framework Matrix

The pipeline validation crosses two domains (SupportLab and OpsLab), two frameworks (LangGraph and AutoGen), and six verifier conditions. B0 accepts a contract-valid diagnosis without a verifier; B1 applies deterministic verification; B2 and B3 exercise shared-context and isolated semantic-verifier interfaces; B4 exercises an isolated cross-model interface; and B5 composes deterministic and isolated semantic interfaces. In this validation, semantic conditions use declared fake providers and therefore carry no semantic- effectiveness interpretation.

The run schedules $2 \times 2 \times 6 = 24$ cells, records four framework-adapter executions and two parity comparisons, joins labels only after the provider phase, and generates metrics, claim gates, paper assets, and a manifest. The artifact explicitly records `fake_evidence=true`, zero provider calls, and zero GPU calls. Repeating the run must produce byte-identical bundles.

6.4 Real-Provider and Risk Protocol

The result-bearing semantic study uses the same frozen traces and diagnoses across B0–B5. Provider-visible inputs exclude labels, mutation metadata, expected findings, split identity, credentials, and hidden reasoning. The primary semantic endpoint is a correct, evidence-supported diagnosis under execution-derived labels; paired bootstrap intervals resample scenario-template clusters and prespecified primary comparisons use Holm correction.

The group-selective study freezes development, calibration, and untouched test groups before outcome analysis. It reports every $(K^G_\lambda, M^G_\lambda)$ pair, simultaneous upper bound, accepted-group minimum, feasibility decision, and selected candidate. Provider failures and contract-invalid outputs remain visible. No value from the fake-provider matrix enters this study's effectiveness or risk calculations.

7 Results

7.1 Controlled Evidence-Contract Results

Table 1 reports every frozen candidate. All 20 valid reports passed deterministic verification. All 16 mutated reports were intercepted, yielding 20/20 valid-report acceptance, 16/16 hard-defect recall, and 0/20 false blocks on this controlled benchmark. The verifier routed repairable selector and grounding defects to NEEDS-EVIDENCE; it routed scope, hash-integrity, and diagnosis-conflict defects directly to REVIEW-REQUIRED.

These results test exact implemented rules against programmed defects. They do not estimate performance on naturally occurring diagnoses and do not measure the optional semantic verifier. Their contribution is narrower: the claim–evidence contract creates an executable boundary that distinguishes valid reports from six classes of known invalid report.

Table 1: Deterministic verification on the 36-candidate frozen benchmark. "NE" denotes NEEDS-EVIDENCE; "HR" denotes REVIEW-REQUIRED.

Candidate family	*n*	NE	HR
Valid, unmodified	20	0	0
Unsupported scope	6	0	6
Invalid selector	2	2	0
Value/hash mismatch	2	0	2
Claim not grounded	2	2	0
Critical span not grounded	2	2	0
Diagnosis conflict	2	0	2
All injected defects	16	6	10

Table 2: SpanVouch engineering validation. The exact public Git revision is stated in the accompanying text.

Validation surface	Observed result
Code snapshot	Public Git revision named above
Public contracts	6 versioned roots with strict valid/invalid fixtures
Release suite	1,637 passed, 1 skipped; 93.40% coverage
Static/delivery gates	Ruff, strict mypy, locked wheel, and Compose passed
Process recovery	20/20 SQLite processes completed
Container recovery	Unprivileged API retained the terminal decision after restart
Frozen artifact	Manifest and metrics hashes reproduced exactly

7.2 System Validation Results

Table 2 summarizes the system-level evidence. The audited public snapshot at Git revision `441871aa19cd4d7c129a721a449c5a098780afd1` collects 1,638 tests; its complete local release gate recorded 1,637 passes, one platform-specific skip, and 93.40% statement coverage. Static analysis, packaging, Compose validation, and the unprivileged container path passed. The accepted recovery gate also completed all 20 independent SQLite stability processes and preserved an end-to-end review decision across an API-container restart.

7.3 Offline Matrix Results

The offline matrix evaluated all 24 scheduled domain–framework–condition cells. It completed four adapter executions and both framework-parity comparisons, then generated the evaluated-results manifest, statistical assets, claim-gate report, and paper assets. Repeated executions produced identical bundles. The run attempted zero provider requests and zero GPU operations.

The matrix result establishes that every B0–B5 delivery and analysis interface is executable across both labs and framework adapters. It does not rank the six conditions: fake semantic-provider outputs intentionally make the recorded condition risks unsuitable for an effectiveness comparison. Real-provider false-acceptance, correlated-error, and empirical target-risk results therefore remain outside the reported findings.

8 Limitations and Threats to Validity

The term "independent" denotes controlled failure separation, not statistical independence of verifier errors. Models may share training data, inductive biases, or misleading trajectory abstractions even when their APIs and prompts differ. Conditioned comparisons can estimate residual error correlation, but cannot prove that all shared failure causes have been removed.

The selective-risk statement is conditional on the frozen loss and pipeline, independently sampled preregistered groups, a finite candidate family, simultaneous correction, and a positive accepted-group minimum. Group-disjoint splitting and grouped bootstrap intervals do not establish independent group sampling. The group-level any-defect loss can also be more conservative than report-level risk. Distribution, provider, and temporal shifts remain outside an in-distribution certificate unless separately calibrated or covered by a justified shift procedure.

Controlled fault injection supplies reproducible causal labels but may not match naturally occurring compound failures. We therefore separate injected, replay-derived, and naturally observed cohorts and avoid pooling them without stratification. A minimal sufficient evidence set is not always unique. Evidence metrics must allow prespecified equivalent sets or adjudicated alternatives rather than penalizing every non-canonical proof. If a critical label remains unresolved after the preregistered pre-calibration adjudication, the primary certificate conservatively assigns loss one. This can overstate risk, but excluding unresolved reports would change the estimand and require a separately preregistered missingness or selection assumption. Alternative label treatments are therefore uncertified sensitivity analyses. Fifth, one bounded read-only action cannot recover evidence that was never logged or that requires an unsafe side effect. Such cases must remain abstentions.

The reported deterministic benchmark tests programmed contract defects, and the offline matrix uses declared fake providers. These results establish implemented rule behavior and pipeline reproducibility, not semantic-verifier effectiveness or empirical target-risk attainment. No human-participant study supports a claim about debugging time or operational decision quality.

9 Conclusion

IVAD frames agent-failure diagnosis as a selective decision over machine-checkable causal claims rather than an unconstrained explanation. It combines re-addressable trajectory evidence, deterministic eligibility, controlled semantic verification, bounded revision and acquisition, and a group-selective acceptance protocol. Under a frozen pipeline, independently sampled preregistered groups, and simultaneous finite-candidate bounds, the protocol provides a calibration-only PAC statement with an explicit no-feasible-point outcome.

SpanVouch realizes this trust boundary as a versioned, recoverable open-source system. On the frozen 36-candidate benchmark, deterministic verification accepted all 20 valid reports and intercepted all 16 injected defects. The audited release gate recorded 1,637 passes, one platform-specific skip, and 93.40% coverage, while the 24-cell offline matrix reproduced the complete multi-domain, multi-framework artifact path with zero provider and GPU calls.

These results establish executable contract behavior, system integrity, and reproducibility. They do not turn fake-provider execution into a semantic effectiveness result or an empirical selective-risk certificate. Together, IVAD and SpanVouch provide a precise protocol and an engineering substrate for auditable acceptance of agent-failure diagnoses.

References

[1] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2024. Conformal Risk Control. In *International Conference on Learning Representations*. https://proceedings.iclr.cc/paper_files/paper/2024/hash/f3549ef9b5ff520a7e41ff3cc306ab2b-Abstract-Conference.html

[2] Tian Bai and Ying Jin. 2026. Conformal Selective Prediction with General Risk Control. *arXiv preprint arXiv:2603.24704* (2026). https://arxiv.org/abs/2603.24704

[3] Shraddha Barke, Arnav Goyal, Alind Khare, Avaljot Singh, Suman Nath, and Chetan Bansal. 2026. AgentRx: Diagnosing AI Agent Failures from Execution Trajectories. *arXiv preprint arXiv:2602.02475* (2026). https://arxiv.org/abs/2602.02475

[4] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. 2025. Why Do Multi-Agent LLM Systems Fail?. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*. https://proceedings.neurips.cc/paper_files/paper/2025/hash/b1041e52d3be19f0a9bc491657488e4a-Abstract-Datasets_and_Benchmarks_Track.html

[5] Mengzhuo Chen, Junjie Wang, Zhe Liu, Yawen Wang, Haiming Zheng, and Qing Wang. 2026. From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint arXiv:2606.06324* (2026). https://arxiv.org/abs/2606.06324

[6] C. J. Clopper and E. S. Pearson. 1934. The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial. *Biometrika* 26, 4 (1934), 404–413. doi:10.1093/biomet/26.4.404

[7] Darshan Deshpande, Varun Gangal, Hersh Mehta, Jitin Krishnan, Anand Kannappan, and Rebecca Qian. 2025. TRAIL: Trace Reasoning and Agentic Issue Localization. *arXiv preprint arXiv:2505.08638* (2025). https://arxiv.org/abs/2505.08638

[8] Naihe Feng, Yi Sui, Shiyi Hou, Ga Wu, and Jesse C. Cresswell. 2026. Conformal Agent Error Attribution. *arXiv preprint arXiv:2605.06788* (2026). https://arxiv.org/abs/2605.06788

[9] Jeffrey Flynt. 2026. GroundEval: A Deterministic Replacement for LLM-as-Judge in Stateful Agent Evaluation. *arXiv preprint arXiv:2606.22737* (2026). https://arxiv.org/abs/2606.22737

[10] Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong Park. 2024. Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, 7036–7050. doi:10.18653/v1/2024.naacl-long.389

[11] Elliot Myunghoon Kim, Avi Garg, Kenny Peng, and Nikhil Garg. 2025. Correlated Errors in Large Language Models. In *Proceedings of the 42nd International Conference on Machine Learning*. https://openreview.net/forum?id=kzYq2hfyHB

[12] Xiaoxi Li, Guanting Dong, Jiajie Jin, Yuyao Zhang, Yujia Zhou, Yutao Zhu, Peitian Zhang, and Zhicheng Dou. 2025. Search-o1: Agentic Search-Enhanced Large Reasoning Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 5420–5438. doi:10.18653/v1/2025.emnlp-main.276

[13] Jiale Liu, Huajun Xi, Shaokun Zhang, Yifan Zeng, Tianwei Yue, Chi Wang, Jian Kang, Qingyun Wu, and Huazheng Wang. 2026. Who&When Pro: Can LLMs Really Attribute Failures in AI Agents? *arXiv preprint arXiv:2607.09996* (2026). https://arxiv.org/abs/2607.09996

[14] Xing Han Lù, Amirhossein Kazemnejad, Nicholas Meade, Arkil Patel, Dongchan Shin, Alejandra Zambrano, Karolina Stańczak, Peter Shaw, Christopher J. Pal, and Siva Reddy. 2025. AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories. *arXiv preprint arXiv:2504.08942* (2025). https://arxiv.org/abs/2504.08942

[15] Ming Ma, Jue Zhang, Fangkai Yang, Yu Kang, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. 2026. DoVer: Intervention-Driven Auto Debugging for LLM Multi-Agent Systems. In *International Conference on Learning Representations*. https://arxiv.org/abs/2512.06749

[16] Nat McAleese, Rai Michael Pokorny, Juan Felipe Ceron Uribe, Evgenia Nitishinskaya, Maja Trebacz, and Jan Leike. 2024. LLM Critics Help Catch LLM Bugs. *arXiv preprint arXiv:2407.00215* (2024). https://arxiv.org/abs/2407.00215

[17] Hezhe Qiao, Hanghang Tong, Ee-Peng Lim, Bing Liu, and Guansong Pang. 2026. VerifyMAS: Hypothesis Verification for Failure Attribution in LLM Multi-Agent Systems. *arXiv preprint arXiv:2605.17467* (2026). https://arxiv.org/abs/2605.17467

[18] Weihang Su, Yichen Tang, Qingyao Ai, Zhijing Wu, and Yiqun Liu. 2024. DRAGIN: Dynamic Retrieval Augmented Generation based on the Real-time Information Needs of Large Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 12991–13013. doi:10.18653/v1/2024.acl-long.702

[19] Jimmy Wang, Thomas P. Zollo, Richard Zemel, and Hongseok Namkoong. 2025. Adaptive Elicitation of Latent Information Using Natural Language. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 65897–65917. https://proceedings.mlr.press/v267/wang25ff.html

[20] Jian Xu, Delu Zeng, John Paisley, and Qibin Zhao. 2026. Look Again Before You Abstain: Budgeted Conformal Evidence Acquisition for Reliable Vision-Language Model. *arXiv preprint arXiv:2606.16667* (2026). https://arxiv.org/abs/2606.16667

[21] Yunpeng Xu, Wenge Guo, and Zhi Wei. 2025. Selective Conformal Risk Control. *arXiv preprint arXiv:2512.12844* (2025). https://arxiv.org/abs/2512.12844

[22] Zijun Yao, Weijian Qi, Liangming Pan, Shulin Cao, Linmei Hu, Weichuan Liu, Lei Hou, and Juanzi Li. 2025. SeaKR: Self-aware Knowledge Retrieval for Adaptive Retrieval Augmented Generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 27022–27043. doi:10.18653/v1/2025.acl-long.1312

[23] Guibin Zhang, Junhao Wang, Junjie Chen, Wangchunshu Zhou, Kun Wang, and Shuicheng Yan. 2025. AgenTracer: Who Is Inducing Failure in the LLM Agentic Systems? *arXiv preprint arXiv:2509.03312* (2025). https://arxiv.org/abs/2509.03312

[24] Shaokun Zhang, Ming Yin, Jieyu Zhang, Jiale Liu, Zhiguang Han, Jingyang Zhang, Beibin Li, Chi Wang, Huazheng Wang, Yiran Chen, and Qingyun Wu. 2025. Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 76583–76599. https://proceedings.mlr.press/v267/zhang25cq.html